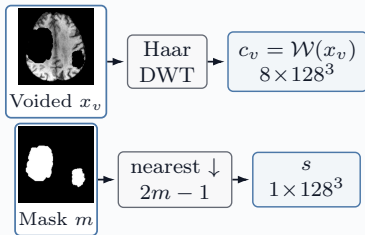


Shared inference-available conditioning and selected fusion F

Condition preparation



Fusion alternatives (one trained per experiment)

Concatenation

$$F_{\text{concat}}(z_t, c_v, s) = [z_t \parallel c_v \parallel s]$$

$8 + 8 + 1 = 17$ channels at 128^3

Spatial input FiLM

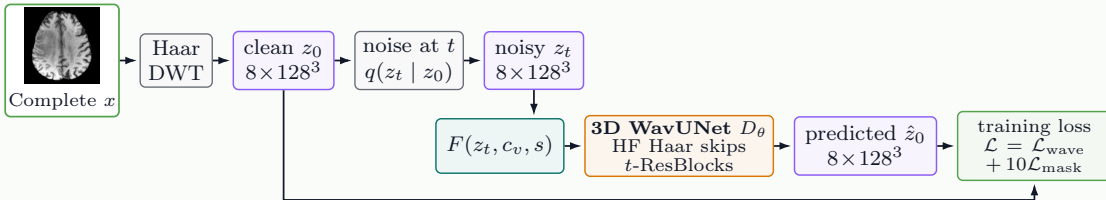
$$[\Delta\gamma \parallel \beta] = C_\phi([c_v \parallel s])$$

C_ϕ : spatial 3D conv encoder

16 spatial maps: $8\Delta\gamma + 8\beta$

$$= (1 + \Delta\gamma) \odot z_t + \beta \quad (8 \text{ channels})$$

(a) Training



(b) Inference: reverse DDPM sampling

Reuse the selected trained fusion module and its corresponding WavUNet weights.

